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T050

Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Dot .Net Core and Progressive Web Application Practical

Practical – 3

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Part B - Program

a) Create methods add(). multiply(), substract() ,divide() with suitable parameters and call these methods using concept of C# delegate.

Algorithm:

Step1 :-Start the program.
step2:-Create a delegate and define calculator methods (Add, Sub, Mul, Div).
step3:-Create an object of the `Calculator` class.
step4:-Assign each method to the delegate and invoke it.
Step5:- Display the results and stop the program.

Code:

```
using System;
delegate void operation(int a, int b);
class Calculator
{
   public void Add(int a, int b)
   {
      Console.WriteLine("Addition =" + (a + b));
   }
   public void Sub(int a, int b)
   {
      Console.WriteLine("Subtraction =" + (a - b));
   }
   public void Mul(int a, int b)
   {
      Console.WriteLine("Multiplication =" + (a * b));
   }
   public void Div(int a, int b)
   {
      Console.WriteLine("Division =" + (a / b));
   }
}
class program
{
   static void Main()
   {
      Calculator c = new Calculator();
      operation obj;
```

```
        obj = c.Add;
        obj(10,10);
        obj = c.Sub;
        obj(20, 10);
        obj = c.Mul;
        obj(20, 10);
        obj = c.Div;
        obj(8, 4);
        Console.WriteLine("Aashu 50");
    }
}
```

Output:

```
Addition =20
Subtraction =10
Multiplication =200
Division =2
Aashu 50
```

b) Write a program using multicast delegate.

Algorithm:

step1**:-**Start the program.

step2:-Create a delegate and define the methods `Welcome()`, `Learn()`, and `Thanks()`.

step3:-Create an object of the `Demo` class.

step4:-Add all methods to the delegate and invoke it.

step5:-Display the messages and stop the program.

Code:

```
using System;
delegate void Message();
class Demo
{
    public void Welcome()
    {
        Console.WriteLine("WELCOME...");
    }
    public void Learn()
    {
        Console.WriteLine("Learning C# Delegates");
    }
    public void Thanks()
    {
        Console.WriteLine("Thank You");
    }
}
public class Program
{
    public static void Main(string[] args)
    {
        Demo d = new Demo();
```

```
        Message m;
        m = d.Welcome;
        m += d.Learn;
        m += d.Thanks;
        m();
        Console.Write("Aashu 50");
    }
}
```

Output:

c) Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

Algorithm:

step1:-Start the program.

step2:-Create the `BankAccount` class with `AccountNumber` and `Balance` properties.

step3:-Create an object of the `BankAccount` class.

step4:-Assign and display the account number and balance.

step5:-Update the balance, display the details, and stop the program.

Code:

```
using System;
class BankAccount
{
    public int AccountNumber { get; set; }
    private double balance;
    public double Balance
    {
        get
        {
            return balance;
        }
        set
        {
            if (balance >= 0)
                balance = value;
            else
                Console.WriteLine("Balance cannot be negative");
        }
    }
    class program
    {
        static void Main()
        {
            BankAccount b = new BankAccount();
```

```
            b.AccountNumber = 110;
            b.Balance = 5000;
            Console.WriteLine("Account Number " + b.AccountNumber);
            Console.WriteLine("Balence" + b.Balance);
            b.Balance = 1000;
            Console.WriteLine("Aashu 50");
        }
    }
}
```

Output:

```
Account number:101
Balance:5000
aashu 50
```